

Advanced Machine Learning for the Physical Sciences

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Two-Part Structure

1 Part I: Discriminative Models

Jan–Mar: NNs, CNNs, RNNs, autoencoders, transformers

2 Part II: Generative Models

Apr–May: Boltzmann machines, VAEs, diffusion, GANs

Unifying Thread

Deep learning mathematics applied directly to problems in the physical sciences.

Methods Covered

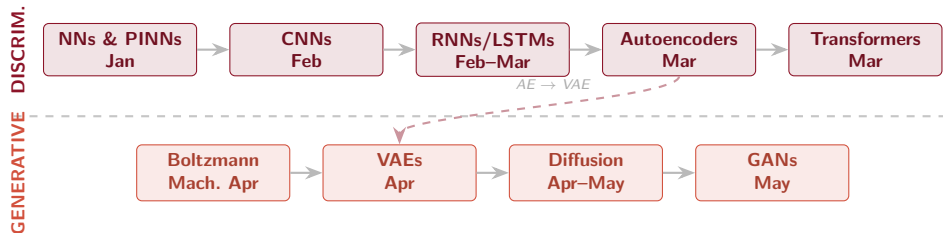
Discriminative

- Feed-forward NNs
- PINNs
- CNNs
- RNNs / LSTMs
- Autoencoders
- Transformers

Generative

- Boltzmann machines
- VAEs
- Diffusion models
- Autoregressive
- GANs

Course Road Map



Part I

Discriminative Deep Learning

January – March

Mathematics of Deep Learning

- Neurons, layers, activation functions
- Forward pass (compact form):
$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$
- Loss functions and optimisation
- Backpropagation: chain rule
- Writing a NN from scratch

Overview & Projects

- Course structure and themes
- Deep learning in the physical sciences: survey
- Discussion of project topics
- Tools: PyTorch / JAX

Goal

Build a NN from first principles before using high-level frameworks.

Deep Learning Mathematics (cont.)

- Optimisers: SGD, Adam, RMSProp
- Regularisation: dropout, batch norm
- Universal approximation theorem
- Practical NN coding session

Solving ODEs / PDEs

- NNs as mesh-free function approximators
- Loss encodes boundary conditions
- Advantages and limitations
- Code examples: simple ODEs

Bridge to the Physical Sciences

NNs that solve differential equations are one of the most direct applications of deep learning to the physical sciences.

PINNs: Key Idea

Embed the PDE directly into the loss function:

$$\mathcal{L} = \mathcal{L}_{\text{PDE}} + \lambda \mathcal{L}_{\text{BC}}$$

- Automatic differentiation for residuals
- Hard vs. soft BC enforcement
- Schrödinger, heat, Poisson as examples

Implementation & Results

- Network architecture choices
- Collocation point sampling
- Training strategies
- Benchmarking vs. classical solvers
- Limitations of PINNs

Week 4: CNN Mathematics

- Discrete convolution and cross-correlation
- Filters, feature maps, receptive fields
- Pooling: max, average, global
- Parameter sharing, equivariance
- Backprop through conv layers

Week 5: Architectures

- Classic architectures: LeNet, ResNet, U-Net
- Skip connections and residual learning
- CNNs for physics: images, fields, spectra

Applications

CNNs excel at identifying spatial patterns in simulation output, detector images, and lattice field configurations.

Recurrent Neural Networks

- Sequential data: time series, trajectories
- Vanilla RNN hidden-state update:
$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$$
- Backpropagation through time (BPTT)
- Vanishing / exploding gradients

Long Short-Term Memory

- LSTM gates: forget, input, output
- Cell state as long-range memory
- GRUs as a lighter alternative
- Applications:
molecular dynamics,
time-series forecasting

Autoencoders

Encoder–decoder: $x \rightarrow z \rightarrow \hat{x}$

- Bottleneck latent space z
- Reconstruction loss $\|x - \hat{x}\|^2$
- Denoising autoencoders
- Convolutional AE architectures

Links to PCA & Implementations

- Linear AE $\equiv$ PCA (exactly)
- Kernel PCA, nonlinear extensions
- Latent space visualisation
- Applications: order parameters, phase classification, compression

The Transformer Architecture

Self-attention: $\text{Att}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$

- Multi-head attention
- Positional encoding
- Encoder / decoder stacks
- Applications: quantum chemistry, PDEs and many other

Part I Summary

Five discriminative families:

- 1 Feed-forward NNs & PINNs
- 2 CNNs
- 3 RNNs / LSTMs
- 4 Autoencoders
- 5 Transformers

Each tackles a distinct structural challenge in data from the physical sciences.

Part II

Generative Deep Learning

April – May

What Are Generative Models?

- Goal: learn $p_{\theta}(x) \approx p_{\text{data}}(x)$
- Explicit vs. implicit density estimation
- Energy-based models: $p_{\theta}(x) = e^{-E_{\theta}(x)} / Z(\theta)$
- Markov Chain Monte Carlo recap
- Gibbs sampling

Boltzmann Machines

- Restricted BM: visible & hidden units
- Contrastive Divergence (CD- k)
- Persistent CD and variants
- Deep Boltzmann Machines
- Statistical mechanics analogy

Energy-Based Models (Deeper Dive)

- Score matching and noise contrastive estimation
- Contrastive Divergence revisited
- Expressivity of energy functions
- Links to partition functions in statistical physics
- Training instabilities and remedies

Boltzmann Machine Codes

- Implementing an RBM from scratch
- Sampling via Gibbs chains
- Visualising learned features
- Wave-function ansatz application
- Benchmarks and diagnostics

Highlight

RBM's have been used as variational ansätze for quantum many-body ground states — a direct link to quantum computing.

VAE: Probabilistic Autoencoders

- Latent variable model: $p(x) = \int p(x|z)p(z) dz$
- ELBO objective: $\mathcal{L} = \mathbb{E}_q[\log p(x|z)] - D_{\text{KL}}(q||p)$
- Reparameterisation trick
- Encoder $\rightarrow (\mu, \sigma)$; Decoder $\rightarrow \hat{x}$

VAE vs. Standard AE

- Structured, continuous latent space
- Sample: $z \sim \mathcal{N}(0, I)$
- Disentanglement (β -VAE)
- Applications:
molecular generation,
anomaly detection,
data interpolation

Denoising Diffusion Probabilistic Models

- Forward: add Gaussian noise gradually
 $q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I)$
- Reverse: learn to denoise step-by-step
- Score matching and Langevin dynamics
- Noise schedules: linear, cosine
- Link to stochastic differential equations

Why Diffusion Models?

- State-of-the-art sample quality
- Stable training vs. GANs
- Conditional generation
- Applications:
molecular structure,
protein folding,
lattice field theory

Diffusion: Codes & Implementations

- Implementing DDPM from scratch
- U-Net backbone for the denoiser
- Training loop and ELBO loss
- Sampling procedures
- Classifier-free guidance

Generative Adversarial Networks

Minimax:

$$\min_G \max_D \mathbb{E}[\log D(x)] + \mathbb{E}[\log(1 - D(G(z)))]$$

- Training dynamics and mode collapse
- Wasserstein GAN (WGAN)
- Conditional GANs (cGAN)

Advanced GAN Topics

- Architectures: DCGAN, StyleGAN, CycleGAN
- Evaluation: FID, IS, precision/recall
- GANs: fast simulation, inverse problems, lattice QCD
- GANs vs. diffusion: trade-offs

Project Presentations

Applications across the curriculum:

- PINNs for PDEs
- CNNs for detector data
- RBMs as wave-function ansätze
- VAEs for molecular design
- Diffusion / GANs for fast simulation

Weekly Road Map — Part I: Discriminative Methods

Week	Dates	Part	Topics
1	Jan 19–23	Discriminative	Course intro, NN basics, backpropagation, project discussion
2	Jan 26–30	Discriminative	NN, optimisers, regularisation, solving ODEs with NNs
3	Feb 2–6	Discriminative	Physics-Informed Neural Networks (PINNs)
4–5	Feb 9–20	Discriminative	Convolutional NNs: mathematics, architectures
6–7	Feb 23–Mar 6	Discriminative	Recurrent NNs, LSTMs, backprop through time
8–9	Mar 9–20	Discriminative	Autoencoders, PCA, latent space, implementations
10	Mar 23–27	Discriminative	Transformers, self-attention, Part I summary

Weekly Road Map — Part II: Generative Methods

Week	Dates	Part	Topics
11	Apr 6–10	Generative	Generative models overview, Boltzmann machines, MCMC, Gibbs
12	Apr 13–17	Generative	Energy-based models, wave-function ansatz
13	Apr 20–24	Generative	Variational autoencoders (VAEs), ELBO, reparameterisation
14	Apr 27–May 1	Generative	Diffusion models: forward/reverse process, score matching
15	May 4–8	Generative	Diffusion codes (DDPM), GANs: minimax, WGAN, cGAN
16	May 18–22	Generative	Advanced GANs, evaluation metrics, project presentations

What You Will Know at the End

Discriminative Skills

- Derive and implement backpropagation
- Build NNs, CNNs, RNNs, and PINNs
- Solve ODEs and PDEs with neural networks
- Design autoencoders for data reduction
- Understand attention and transformers

Generative Skills

- Train Boltzmann machines and sample from them
- Derive and implement the VAE ELBO
- Implement denoising diffusion models
- Train and stabilise GANs
- Apply generative models to problems in the physical sciences

Overarching Goal

Build both the **mathematical foundations** and the **practical implementation skills** to apply state-of-the-art deep learning methods to open problems in the physical sciences.

Perspectives

Future Directions in Advanced ML

For the Physical Sciences and Beyond

Physics

- **Many-body quantum systems:** neural-network quantum states (NQS) as variational ansätze for ground and excited states
- **High-energy physics:** jet tagging, event generation, fast detector simulation with GANs and diffusion models
- **Astro- and cosmological data:** field-level inference, gravitational-wave denoising, CMB analysis with score-based generative models
- **Condensed matter:** phase classification with CNNs, ML interatomic potentials for MD

Chemistry

- **Molecular generation:** VAEs, diffusion models, and flow-matching for de-novo drug and catalyst design
- **Property prediction:** graph neural networks (GNNs) on molecular graphs: energies, spectra, reaction rates
- **Quantum chemistry:** ML surrogates for DFT and CCSD(T); PINNs for the electronic Schrödinger equation
- **Reaction pathways:** transformers for retrosynthesis and transition-state prediction

Mathematics

- **Operator learning:** DeepONet and Fourier Neural Operators for mesh-free, resolution-independent PDE solvers
- **Inverse problems:** posterior sampling with score-based diffusion; learned regularisers replacing hand-crafted priors
- **Symbolic regression:** discovering equations and conservation laws directly from data
- **Uncertainty quantification:** Bayesian deep learning and conformal prediction for rigorous error bounds

Geoscience

- **Weather and climate:** data-driven emulators (GraphCast, FourCastNet) for fast global forecasts; hybrid physics–ML climate models
- **Seismology:** CNNs and RNNs for earthquake detection, waveform inversion, and induced-seismicity monitoring
- **Remote sensing:** multi-modal transformers for land use, ice-sheet dynamics, and ocean surface reconstruction
- **Subsurface imaging:** generative models for reservoir characterisation and seismic interpretation

Structural and Molecular Biology

- **Protein structure prediction:** AlphaFold2 and successors combine transformers with equivariant GNNs; diffusion models (RFDiffusion) for *de-novo* protein design
- **RNA and DNA:** sequence-to-function models, splicing prediction, epigenomics
- **Drug discovery:** generative molecular design, binding affinity prediction, multi-target optimisation

Systems and Medical Biology

- **Single-cell genomics:** VAEs and diffusion for cell-type discovery and trajectory inference
- **Medical imaging:** segmentation and anomaly detection with CNNs and score-based models; synthetic data for rare-disease augmentation
- **Biophysical simulations:** ML force fields (ANI, NequIP) and coarse-grained MD for proteins and membranes

Reinforcement Learning for the Physical Sciences

- **Quantum control:** RL agents learn pulse sequences for high-fidelity gate synthesis and robust quantum error correction
- **Experiment design:** active learning with RL for adaptive data acquisition in particle physics and materials science
- **Molecular design:** RL-driven exploration of chemical space guided by property reward functions
- **Plasma and fusion:** real-time magnetic confinement via deep RL policies (DeepMind / ITER)

Agentic AI in Science

- **Autonomous research loops:** LLM-based agents that formulate hypotheses, run simulations, interpret results, and iterate without human intervention
- **Multi-agent systems:** specialised agents for literature search, code generation, data analysis, and experimental planning collaborate in a shared scientific workflow
- **Human-in-the-loop science:** agentic pipelines as co-pilots for accelerating discovery in drug design, materials screening, and climate modelling

Graph Neural Networks

- Natural representation of molecules, crystals, interaction networks, and complex systems
- **Equivariant GNNs** (SE(3)-Transformers, NequIP, MACE) enforce physical symmetries exactly
- Applications: molecular force fields, particle-physics event reconstruction, social and biological networks
- **Open challenge:** scaling equivariant architectures to very large systems efficiently

Foundation Models and Transfer Learning

- Large pre-trained models (ESM-2, ChemBERTa, ClimaX, AstroCLIP) transfer to downstream science tasks with little labelled data
- **Scientific LLMs:** models fine-tuned on papers, code, and experimental data for literature synthesis and hypothesis generation
- **Multi-modal learning:** joint text, structure, spectrum, and image representations for universal scientific models

Outlook

The frontier: models that are simultaneously **data-efficient**, **physically interpretable**, and **generalisable** across scientific disciplines.